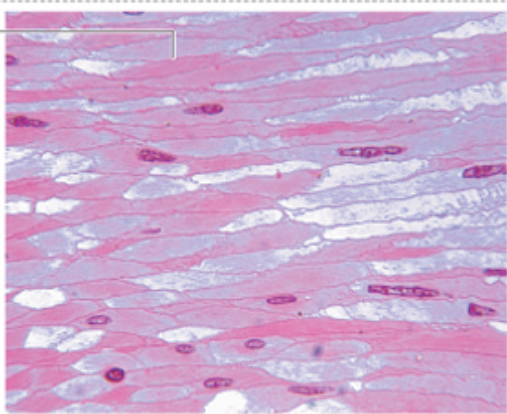
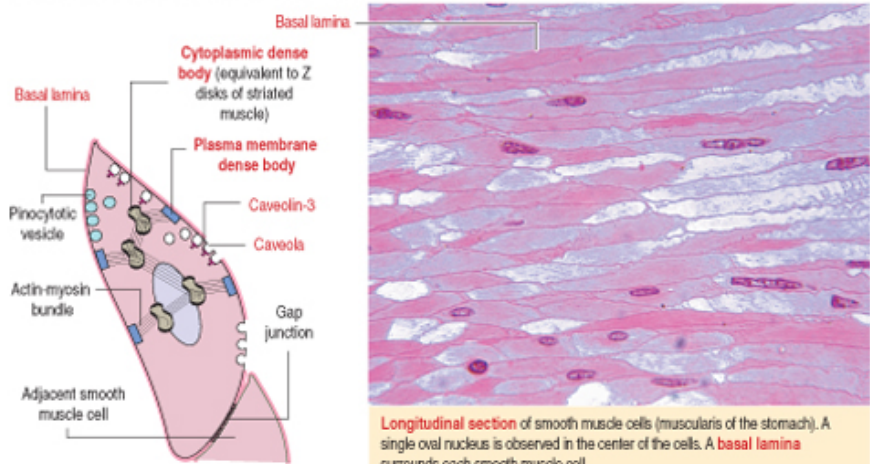


SMOOTH MUSCLE

- Spindle-shaped cells
- Length from 20 to 200 μm
- One nucleus at the center
- The smooth muscle cells are able to synthesize elastin, collagen and other extracellular matrix's components
- No striation
- Regenerable

SMOOTH MUSCLE



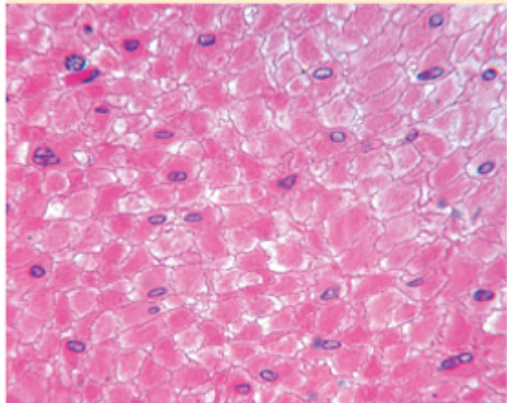
Longitudinal section of smooth muscle cells (muscularis of the stomach). A single oval nucleus is observed in the center of the cells. A basal lamina surrounds each smooth muscle cell.

Characteristics of smooth muscle

Smooth muscle is found in the walls of tubular organs, the walls of most **blood vessels**, the **iris** and **ciliary body** (eye), and **arrector pili muscle** (hair follicles), among other sites. It consists of fusiform individual cells or fibers with a **central nucleus**. Smooth cells in the walls of large blood vessels produce **elastin**.

Caveolae—depressions of the plasma membrane—are permanent structures involved in fluid and electrolyte transport (**pinocytosis**).

Caveolin-3, a protein encoded by a member of the caveolin gene family, is associated with **lipid rafts**. Complexes formed by caveolin-3 bound to **cholesterol** in a lipid raft invaginate and form caveolae. Caveolae detach from the plasma membrane to form **pinocytotic vesicles**.



Cross section of smooth muscle cells. Depending on the section level, a central nucleus is observed in some of the muscle cells.

